MILITARY SPECIFICATION

IGNITER, JET ENGINE, MK 243 MOD 2

1. SCOPE

1.1 This specification governs the manufacture, assembly and preparation for delivery of the Igniter, Jet Engine Mk 243 Mod 2 and methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood-Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lockcorner
PPP-B-636	Boxes; Fiber
PPP-B-676	Boxes; Set-up, Paperboard

FSC 1375

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Military

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material; Water Vaporproof, Flexible

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-350	Jolt Test for use in Production of Fuzes
MIL-STD-351	Jumble Test for use in Production of Fuzes
MIL-STD-352	Forty (40) Foot Drop Test for use in Production of Fuzes
MIL-STD-354	Temperature and Humidity Test for Use in Production of Fuzes

DRAWINGS

Bureau of Ordnance

LD 282766	Igniter, Jet Engine, Mk 243 Mod 2

and all drawings, standards and specifications listed on the above LD.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

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2.2 Other publications. The following publication forms a part of this document. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite '49 CFR 71-78' latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General

3.1.1 The Igniter, Jet Engine Mk 243 Mod 2 covered by this specification shall be manufactured in accordance with LD 282766 and all documents listed thereon.

3.1.2 Definition. The term igniter as used herein shall mean the Igniter, Jet Engine Mk 243 Mod 2.

3.1.3 Preproduction lot. Unless otherwise specified for each contract or order, a representative sample of 200 igniters shall be furnished by the contractor at the contractor's expense. These igniters shall be manufactured using the methods and procedures proposed for the production lot and shall meet all requirements specified herein.

3.1.4 Production lot. For procurement purposes each production lot of igniters shall consist of 5,000 units plus those required for all test purposes. Unless otherwise stated in contract, order or requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.2.1.

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3.1.5 Radiographs. Prior to Government testing, the contractor shall furnish the Government inspector, at the contractor's expense, with a radiograph of the preproduction samples and samples selected for product acceptance of each lot. Each radiograph shall contain the following identification marking:

(a) Lot number
(b) Date of manufacture
(c) Contract number
(d) Manufacturer's name

The radiographs shall be forwarded by the Government inspector to the Quality Evaluation Laboratory, U. S. Naval Ammunition Depot, Crane, Indiana.

3.2 Performance and production characteristics

3.2.1 Jumble. The igniter shall withstand the jumble test prescribed in 4.4.1.

3.2.2 Functional. The igniters, when fired as specified in 4.4.2 at any ambient temperature between minus 65°F and plus 160°F shall function by:

(a) Functioning within one second after actuation of the firing pin of test chamber.
(b) Burn for a period of 12 $\pm$ 3 seconds

No igniter shall misfire.

3.3 Workmanship. Care shall be exercised in the assembly of the component parts to assure proper seating and alignment especially where crimping is employed to maintain parts in their relative positions. There shall be no burrs or other imperfections of manufacture that will affect the assembly or functioning of the product. All parts shall be free from chips, dirt and other foreign materials. The cleaning agent and cleaning methods used shall not be injurious to any of the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements

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prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in determination of acceptability of products submitted by a supplier for the Government shall be in accordance with the provisions of Standard MIL-STD-105.

4.2.1 Lot. As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in inspection lots shall be as determined by the inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.3 Examination

4.3.1 Components. The Government inspector shall acertain that prior to assembly all components of the igniter procured under separate specfications or drawings have been inspected, tested and accepted in accordance with their respective specifications or drawings.

4.3.2 Packing and marking. The Government inspector shall ascertain that the packing and marking of the igniters conform to this specification.

4.4 Acceptance tests. Unless otherwise specified all inspection and tests shall be at the place of manufacture.

4.4.1 Jumble test. Four igniters shall be randomly selected from each seventy-five (75) production samples or multiples thereof selected in accordance with 4.4.2. These igniters shall be subjected to the jumble test of Standard MIL-STD-351.

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4.4.1.1 Acceptance criteria

(1) No explosive or pyrotechnic element shall be initiated.
(2) The igniter shall show no signs of breakdown or loosening which will make the assembly unsafe to handle or dangerous to use.

The failure of one or more igniters to meet the above requirements shall be cause for rejection of the lot. These samples shall be subjected to the functional test of 4.4.2. Failure of one or more igniters to meet the requirements of 3.2.2 shall be cause for rejection of the lot represented.

4.4.2 Functional test. The igniter shall be mounted, axis vertically with the primer end up, in a suitable safety chamber which simulates the emergency igniter combustion chamber of Aircraft Jet Engine J-48. Head space, firing pin protrusion, firing pin tension, firing pin contour, and seating flange diameter shall conform to applicable combustion chamber drawings. The igniter shall be fired by actuating the firing pin. Sampling shall be conducted in accordance with Standard MIL-STD-105, Inspection Level I, AQL 0.65 percent defective. See 6.2 for combustion chamber information.

4.4.2.1 Acceptance. Conformance with requirements of 3.2.2 shall determine acceptability.

4.5 Preproduction lot tests. These tests shall be conducted on the preproduction lot and on such production material as specifically designated by the Government. The preproduction lot must prove acceptable prior to any further production under the contract.

4.5.1 Jumble test. Twenty-four igniters shall be subjected to the jumble test of 4.4.1.

4.5.1.1 Acceptance criteria

(1) No explosive or pyrotechnic element shall be initiated.
(2) The igniter shall show no signs of breakdown or loosening which will make the assembly unsafe to handle or dangerous to use.

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Failure of one or more igniters to meet the above requirements shall be cause for rejection of the lot represented.

4.5.1.1.1 The twenty-four igniters, accepted in accordance with 4.5.1.1, shall be subjected to the functional test of 4.4.2. Failure of two or more igniters to meet requirements of 3.2.2 shall be cause for rejection of the lot represented.

4.5.2 Temperature and humidity test. Fifty igniters shall be subjected to the temperature and humidity test of Standard MIL-STD-354.

4.5.2.1 Acceptance criteria. Following this test the igniters shall be subjected to the functional test of 4.4.2. Failure of four or more igniters to meet the requirements of 3.2.2 shall be cause for rejection of the lot represented.

4.5.3 Forty Foot Drop test. Ten igniters shall be subjected to the forty foot drop test as prescribed in Standard MIL-STD-352. Two igniters shall be subjected to a guided drop; one to strike flange down, and one flange up. The remaining igniters shall be dropped for random impact.

4.5.3.1 Acceptance criteria

(1) The igniter shall be safe to handle and dispose of
(2) No explosive or pyrotechnic element shall be initiated.

Failure of one or more igniters to meet the above requirements shall be cause for rejection of the lot represented.

4.5.4 Jolt test. Ten igniters shall be subjected to the jolt test of Standard MIL-STD-350. Each igniter shall be mounted in a fixture which rigidly supports the igniter by the flanged end and shall be jolted in each of the following positions:

(1) Axis vertical; flanged end up
(2) Axis vertical; flanged end down
(3) Axis horizontal

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4.5.4.1 Acceptance criteria

(1) No explosive or pyrotechnic element shall be initiated.
(2) The igniter shall be safe to handle and dispose of

Failure of one or more igniters to meet the above requirements shall be cause for rejection of the lot represented.

4.5.5 Functional test at high and low temperatures. Fifty igniters shall be subjected to the functional test of 4.4.2; twenty-five igniters shall be tested at minus 65 degrees F ± 5 degrees F and twenty-five igniters shall be tested at plus 160 degrees F ± 5 degrees F. The igniter shall function by:

(a) Functioning within one second after actuation of the firing pin of test chamber
(b) Burn for a period of 12 ± 3 seconds

No igniter shall misfire.

4.5.5.1 Acceptance criteria. Failure of two or more igniters to meet the above requirements shall be cause for rejection of the lot represented.

4.6 Examination. The preproduction lot shall conform to 4.3.1 and 4.3.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.1.1.1 Cleaning, drying, and preservation application. Not applicable.

5.1.1.2 Unit packaging. Ten igniters shall be packaged in accordance with Specification MIL-P-116, Method 1A-15. The inner container shall be a folding carton of set-up box, with separators, conforming to Specification PPP-B-566 or PPP-B-676.

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The heat sealed bag, shall be fabricated from water vapor proof barrier material conforming to Specification MIL-B-131.

5.1.1.3 Intermediate package. Ten unit packages as described in 5.1.1.2 shall be intermediately packaged in a fiberboard box conforming to Specification PPP-B-636, Type I or II, Class 2, Style RSC. Seams, joints and closures shall be taped with material conforming to Specification PPP-T-60.

5.1.2 Level B. Not applicable

5.1.3 Level C. Not applicable

5.2 Packing

5.2.1 Level A. In addition to the following Level A packing shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.2.1.1 Ten intermediate packages as described in 5.1.1.3 shall be packed in a wood cleated plywood box, overseas type conforming to Specification PPP-B-601 or nailed wood box, Class 2 conforming to Specification PPP-B-621. Strapping of the container shall be in accordance with the Appendix of the applicable specification.

5.2.2 Level B. Not applicable

5.2.3 Level C. Not applicable

5.3 Marking

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations '49 CFR 71-78'.

5.3.2 Normal markings. In addition to the markings required by the contract or order. Unit packaged and shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.4 Manufacturing data cards. Five copies of pyrotechnic manufacturing data cards (NavOrd Form 1204) shall be filled out for each lot. The Government inspector shall fill in the

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acceptance data section, endorse the cards, and forward to the procuring agency. The following data shall be listed:

Shell Case	Dwg No.
Primer	Type
Pyrotechnic Composition	Dwg No.
F33-B Igniter	Dwg No.

5.5 Performance data cards. Five pyrotechnic ammunition performance data cards (NavOrd Form 1187) shall be filled out and enclosed in each container. The elements of performance shall be listed as follows:

(a) Primer action (specify time from actuation of the firing pin to functioning)
(b) Burning time

6. NOTES

6.1 Intended use. The Igniter, Jet Engine Mk 243 Mod 2 is for use for the emergency ignition of Aircraft Jet Engine J-48.

6.2 Emergency igniter combustion chamber. The Emergency Igniter Combustion Chamber for Jet Engine J-48 is described in Pratt and Whitney J-48 Turbo Jet Engine Bulletin No. 48, dated 21 August 1954. Other test chambers must be demonstrated to be satisfactory for the purpose intended.

6.3 Ordering data. Procurement documents shall specify:

(a) Title, number and date of this specification
(b) Agency where preproduction samples should be forwarded

6.4 General safety precautions. The loading, assembly and handling of the item covered by this specification, and the sub-assemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions

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escribed atmospheric conditions at all times so that all areas are exposed to the pr

which a contractor or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Bureau of Ordnance, Navy Department, and other government plants, these provisions are mandatory. Such other warnings and precautions pertinent to the operational effectiveness or safety during use or loading of the specified item should be included in the detail technical requirements of the specification.

6.4.1 All loading operations should be conducted in a neat and orderly manner.

6.4.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, coating, crimping, etc.

6.4.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

6.4.4 The exposure of explosive materials and related parts shall be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.4.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards, and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intra-plant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.4.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

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6.4.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.4.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.4.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.4.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures; and where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.4.6.5 Protect all explosive operations from effects or electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may

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be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.4.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.4.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.4.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping and equipment or the items loaded; also, clean up any spilled material immediately.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the U. S. Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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